Structural analogies as posterior transfer in continual reinforcement learning: When does analogy help?

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Abstract

Biological agents continuously draw on past experiences to solve novel problems by analogy – a capacity widely regarded as a hallmark of intelligence (Gentner 1983; Shanahan and Mitchell 2022). Reinforcement learning (RL) provides the framework to study such agents; yet RL remains sample inefficient and struggles to generalize beyond its training distribution (Lake et al. 2017; Shanahan and Mitchell 2022), potentially due to a failure to draw analogies and thereby exploit earlier experiences (Shanahan and Mitchell 2022). We formalize analogy-making for continual model-based RL as the online detection of recurring transition structures within a growing state space and define analogies as structural priors. We show that analogy-making agents significantly outperform agents without analogy and that transfer is beneficial when the error induced by the analogy is smaller than the local sampling error. In particular, performance improves whenever agents have many similar experiences from which to draw analogies. Our results suggest that enabling RL agents to make analogies is a path toward the sample-efficient, adaptive behavior observed in biological agents.

Introduction

A central challenge in reinforcement learning (RL) is sample efficiency, as agents may require many interactions with the environment before learning useful behavior. One route to efficiency is exploration: optimistic methods and posterior-sampling methods use uncertainty to improve sample efficiency (Kearns and Singh 2002; Brafman and Tennenholtz 2002; Asmuth et al. 2009; Osband, Russo, and Van Roy 2013). Posterior-sampling RL (PSRL; Osband, Russo, and Van Roy 2013) maintains a Bayesian posterior over environment dynamics, samples a single MDP at each episode, and plans optimally on it, using uncertainty to drive exploration implicitly. A complementary approach to exploration is structural generalization: abstraction and symmetry-based methods share experience across states by grouping them under behavioral or transformation-based invariances (Ravindran 2004; Ferns, Panangaden, and Precup 2011; Abel et al. 2018; Van der Pol et al. 2020; Wang, Walters, and Platt 2022). This can accelerate learning by reducing the effective state or policy space. Building on both routes, we propose an agent that uses structural correspondences between observed regions in state space to define structural priors over unvisited regions, thereby guiding exploration by structural similarity rather than uncertainty alone.

Despite its potential to improve sample-efficienty, the online discovery of structurally similar regions to support analogical transfer has received little formal treatment in RL. We address this gap by formalizing *analogy-making*: a transfer operation in which transition statistics from visited regions define a structural prior over the dynamics of an under-explored region. This kind of analogy-making operation is useful when local structures repeat, such as floors with similar layouts. However, correspondences are usually approximate: two regions may be similar but not identical. This raises the question when is analogy-making beneficial, and how does the degree of approximation affect the quality of transfer?

Existing approaches detect similarity only among nearby visited states (Mandel et al. 2016), or assume the symmetry relation holds globally across the MDP, with no mechanism to quantify how well it holds between specific regions (Van der Pol et al. 2020). This leaves the online discovery of structural correspondences to distant unvisited regions and the conditions under which they support reliable transfer unaddressed. Our contributions are: (i) we formalize analogy-making in continual model-based RL as prior transfer through structural correspondences; (ii) we derive a posterior-error decomposition showing how transfer error separates into analogy error and local sampling error; and (iii) we implement a graph-matching PSRL agent and show that it accelerates learning when repeated structure is sufficiently well aligned.

Preliminaries

We model the environment as a discrete Markov decision process (MDP) $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ the action space, $P(s' \mid s, a)$ the unknown tran-

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sition dynamics, R the reward function, and $\gamma \in [0, 1)$ the discount factor. We assume a factored state space $\mathcal{S} = \mathcal{S}_1 \times \cdots \times \mathcal{S}_k$, where each factor $\mathcal{S}_i$ represents an independent dimension of the state. The state space $\mathcal{S}$ is unknown and discovered online.

A continual agent solves a sequence of goals within a budget of T interaction steps, inducing an MDP $\mathcal{M}_g = (\mathcal{S}, \mathcal{A}, P, R_g, \gamma)$, where only the reward function R_g changes while P remains fixed. The agent seeks to maximize the undiscounted cumulative reward $J = \sum_{t=0}^{T-1} r_t$ across goals without observing g. Rewards are binary and sparse ($r = 1$ only upon goal completion), so this objective is equivalent to maximizing the number of goals solved, which, in turn, depends on learning P efficiently.

When are analogies beneficial?

Analogy mismatch. Many environments contain regions with approximately invariant transition structure (e.g., floors with similar layouts), which – if detected – can be used as analogies. If one or more of these regions has been explored well, its observed dynamics can be used as a prior for an under-explored region. Whether this is beneficial will depend on the discrepancy between the under-explored region dynamics and the transported transition distribution built from previously observed regions (i.e., on the validity of the analogy).

Let $Q_{s,a} = P_{\psi(s),a}$ denote the transported transition distribution, where $\psi : \mathcal{C} \to \mathcal{C}'$ maps states in the under-explored region $\mathcal{C}$ to corresponding states in the analogy $\mathcal{C}'$. We formalize the quality of a correspondence via an *analogy discrepancy* $D(\mathcal{C})$ over an under-explored region $\mathcal{C} \subseteq \mathcal{S}$, defined as the average total-variation (TV) distance between the true transition distribution P and the transported transition distribution Q

$$D(\mathcal{C}) \triangleq \frac{1}{|\mathcal{C}||\mathcal{A}|} \sum_{s \in \mathcal{C},\, a \in \mathcal{A}} d_{s,a}, \qquad d_{s,a} = \left\| P_{s,a} - Q_{s,a} \right\|_{\mathrm{TV}}. \tag{1}$$

We call $d_{s,a}$ the *analogy mismatch* for a state–action pair. $D = 0$ indicates exact transition invariance between the under-explored region and the transported distribution; larger D indicates poorer analogy.

Finding analogies. We present an agent (PSRL-GM) that builds a state-transition graph online, partitions the *visited* state space into regions, and matches the current, under-explored region to a previously explored one via graph matching (Kuhn 1955). The variant of graph matching we use exploits both the observed transition structure and geometric similarities in the factored MDP (e.g., translations along state factors). Because experience in the current region is limited, it may be uncertain which of the previously explored regions is truly the best match. Therefore, the matched region is compared to $M - 1$ additional reference regions, and their transition statistics are transported via their respective correspondences ψ and aggregated into a single prior estimate $\hat{\alpha}_{s,a}$. The resulting analogy is encoded as a Dirichlet prior $\mathrm{Dir}(\lambda \hat{\alpha}_{s,a})$ over successor states, where $\hat{\alpha}_{s,a}$ is the mean of the aggregated transition distribution (estimated from $M \geq 1$ reference regions – when $M = 1$, only the single matched region is used – via Minka's fixed-point algorithm (Minka 2000)), and the fixed $\lambda \geq 0$ scales this mean into the Dirichlet concentration parameters, controlling the prior's strength relative to the observed data.

Posterior error bound. The prior generated from analogies is combined with experience to define a posterior over the environment's transition structure. We now characterize when this posterior is more accurate than an analogy-free, empirical transition model. Let $\tilde{P}_{s,a}$ denote the posterior mean transition distribution at a state–action pair (s, a) after $n_{s,a}$ local observations, and let $P_{s,a}$ denote the true distribution over the same successor states. With probability at least $1 - 2\eta$, the posterior error is bounded as

$$\|\tilde{P}_{s,a} - P_{s,a}\|_{\mathrm{TV}} \leq \underbrace{\frac{\lambda}{\lambda + n_{s,a}}}_{\text{prior weight}} \Bigg[\underbrace{\rho_{s,a}(\eta)}_{\text{anlg. sampling bound}} + \underbrace{d_{s,a}}_{\text{anlg. mismatch}} \Bigg] + \underbrace{\frac{n_{s,a}}{\lambda + n_{s,a}}}_{\text{local weight}} \underbrace{r_{s,a}(\eta)}_{\text{local sampling bound}}. \tag{2}$$

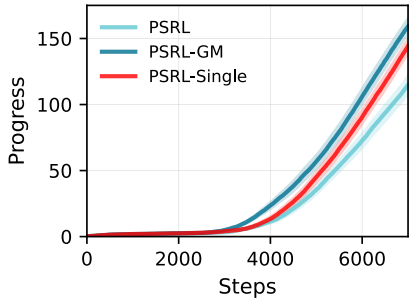
The bound is a convex combination of two terms: the error induced by the analogy and a bound on the empirical error ($\|\hat{P}_{s,a} - P_{s,a}\|_{\mathrm{TV}}$, where $\hat{P}$ is the empirical transition estimate) between the observations and the true transition distribution. The analogy error comes from two sources: (i) the analogy sampling bound $\rho_{s,a}(\eta)$, a high probability bound (at confidence $1-\eta$) on $\|\hat{\alpha}_{s,a} - Q_{s,a}\|_{\mathrm{TV}}$, introduced by accumulating prior information over a finite number of M potentially under-explored matched regions, and (ii) the analogy mismatch $d_{s,a} = \|P_{s,a} - Q_{s,a}\|_{\mathrm{TV}}$, which captures potential structural mismatches between the analogy and the under-explored region to which it is applied. Both TV errors scale as $O(1/\sqrt{N_{\mathrm{total}}})$ and $O(1/\sqrt{n_{s,a}})$, respectively, up to support size and confidence factors (Weissman et al. 2003), where N_{total} denotes the pooled sample size for estimating $Q_{s,a}$, and $n_{s,a}$ is the transition count for (s, a). They decline as N_{total} or $n_{s,a}$ grows, e.g., when additional matched regions add observation counts.

Proof sketch. Subtract the true transition distribution from the convex-combination form of the posterior mean and apply the triangle inequality in TV distance. The resulting analogy error, $\|\hat{\alpha}_{s,a} - P_{s,a}\|_{\mathrm{TV}}$, is bounded by $\rho_{s,a}(\eta) + d_{s,a}$, while the empirical local error $\|\hat{P}_{s,a} - P_{s,a}\|_{\mathrm{TV}}$ is bounded by $r_{s,a}(\eta)$.

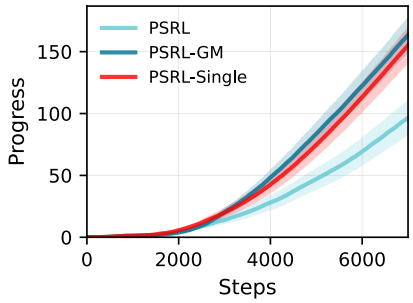
The main conclusion is simple: transfer is *beneficial* whenever $\rho_{s,a}(\eta) + d_{s,a}$ is smaller than the local sampling bound $r_{s,a}(\eta)$. As $n_{s,a}$ grows, local observations down-weight the prior, allowing recovery from mismatch. The simulation lemma (Kearns and Singh 2002) bounds value error in terms of model error, relating transition-model accuracy to planning performance.

Key results

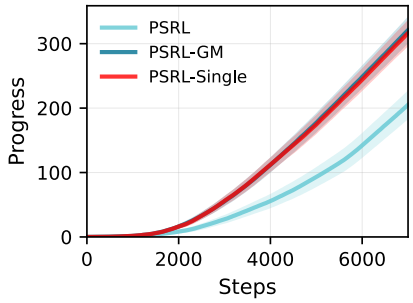
Structural priors help posterior-sampling agents. We evaluate our online analogy-making agent (PSRL-GM) in three stochastic domains with different forms of repeated structure: (i) **SequentialFetch**, ordered item collection as in MiniGrid/BabyAI-style goals (Chevalier-Boisvert et al. 2023, 2018); (ii) **MultiFloorOffice**, office MDP extended to multiple floors via staircases or elevators (Evans and Şimşek 2023); and (iii) **Taxi**, 7×7 Taxi with the passenger's destination treated as the goal (Dietterich 2000). These domains vary in correspondence quality: in SequentialFetch, $D(\mathcal{C}) \approx 0$ for intended correspondences; in MultiFloorOffice, walls are shifted across floors; and in Taxi, the transition structure changes when carrying the passenger (see Appendix A for an illustration).



(a) SequentialFetch progress



(b) MultiFloorOffice progress



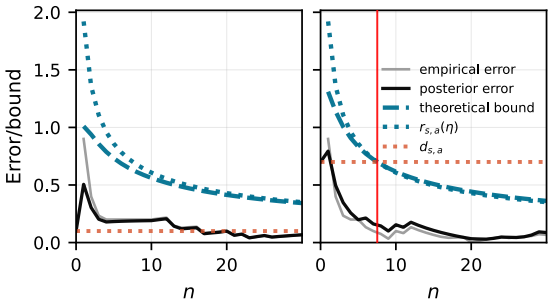
(c) Taxi progress

Figure 1: **Analogies improve continual learning.** Panels (a–c) show the mean over 50 seeds; shaded regions indicate SEM. PSRL-GM outperforms PSRL in all our domains.

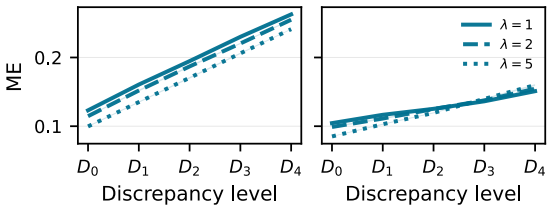
In all three experiments, goals are sequentially sampled from a fixed distribution over $T = 7000$ environment steps. We measure progress as the number of goals solved up to a given time, averaged across seeds, and compare PSRL-GM and PSRL-Single ($M = 1$) to PSRL. All use a PSRL implementation extended to handle a growing state space by maintaining a transition probability to a dedicated exploration state that is inversely proportional to local visit counts, related to the optimistic pseudo-state of Tiapkin et al. (2022).

PSRL-GM improves progress throughout (Fig. 1), showing that structural analogies accelerate learning when useful correspondences exist. Analogies built from multiple consistent reference regions further improve performance (Fig. 1a, 1b). When fewer reference regions exist, the benefit of $M > 1$ is less apparent (Fig. 1c).

The benefit of transfer depends on structure mismatch. Assessing transfer quality is difficult in a continuously growing state space model. To nevertheless evaluate posterior accuracy and model error, we provide agents with oracle access to the transition statistics of reference regions and fix state correspondences manually from $t = 0$, bypassing the graph matching and analogy finding steps. We evaluate our analogy-making agent in the Office domain – a grid-world with four walls surrounding a central open space.



(a) Left: Small $d_{s,a}$. Right: large $d_{s,a}$.



(b) Left: Early ME ($t = 0$). Right: Late ME ($t = 20000$).

Figure 2: **The quality of transfer.** (a) Posterior error vs. visits $n_{s,a}$ for two state–action pairs adjacent to a shifted wall, with small and large $d_{s,a}$: posterior error (black), empirical error (grey), theoretical bound (dashed), local sampling bound $r_{s,a}(\eta = 0.01)$ (dotted). The horizontal line marks $d_{s,a}$; the red vertical line marks $d_{s,a} = r_{s,a}(\eta)$. (b) At $t = 0$, model error (Eq. 3) increases with discrepancy D; larger λ reduces posterior-sampling variability. At $t = 20000$, local observations shrink the benefit of large λ; for larger D, stronger priors preserve mismatch longer.

We define state correspondences from a reference region with one shifted wall, inducing an analogy mismatch $d_{s,a} > 0$ for state–action pairs adjacent to the wall. We select two representative state–action pairs and plot quantities derived from Eq. 2 and the empirical error as a function of visit counts $n_{s,a}$ (Fig. 2a). For both state–action pairs, the posterior error starts at the analogy mismatch $d_{s,a}$, corresponding to the prior with no observation data, and generally decreases as observations accumulate (Fig. 2a). For a small analogy mismatch, the empirical error is slightly above the posterior error, confirming beneficial transfer (Fig. 2a (left)). However, for a large analogy mismatch, the empirical error falls below the posterior error after a few counts, and the condition $d_{s,a} > r_{s,a}(\eta)$ is met soon after, together confirming negative transfer (Fig. 2a (right)). These cases confirm that the crossover condition $d_{s,a} = r_{s,a}(\eta)$ correctly predicts when analogical transfer helps and when it hurts, identifying the crossover after only a few observations.

Having validated the bound at the state–action level, we now ask whether correspondence quality controls model error globally. We sample goals continuously in the same MDP for $T = 20000$ interaction steps. We repeat the experiment with five levels of discrepancy (Eq. 1; we drop M for clarity), with $D_0 = 0 < D_1 < D_2 < D_3 < D_4$. We measure model error over visited state–action pairs,

$$\mathrm{ME}(t) \triangleq \frac{1}{|\mathcal{V}_t|} \sum_{(s,a)\in\mathcal{V}_t} \mathrm{TV}\big(P^*(\cdot \mid s,a), \hat{P}_t(\cdot \mid s,a)\big), \quad \mathcal{V}_t = \{(s,a) : N_t(s,a) > 0\}. \tag{3}$$

At early stages ($t = 0$), we find that $\mathrm{ME}(t)$ increases approximately linearly with D, slopes are determined solely by D, and larger λ reduces posterior sampling variability (Fig. 2b (left)). For late stages of learning ($t = 20000$), the effect of λ dominates: larger λ benefits agents with small D, while for large D, stronger priors preserve mismatch longer (Fig. 2b (right)). These results confirm that transfer quality depends on analogy mismatch and that the effects of mismatched priors are attenuated with visits – although more slowly for larger λ.

Discussion

We presented a framework that formalizes a core claim of structure mapping theory (Gentner 1983): the quality of analogical transfer depends on the degree of structural similarity. In continual model-based RL, this manifests as a tradeoff between the analogy error and local sampling error. Our bound provides a formal criterion for when analogy helps: transfer improves posterior estimates whenever the analogy error is smaller than the local sampling error.

Because the structural prior is weighted by $\lambda/(\lambda + n_{s,a})$, analogy mainly matters when observations are scarce; local observations eventually dominate the prior, correcting for analogy mismatch. The decomposition also explains when defining priors from multiple reference regions ($M > 1$) can be better than relying on a single reference region. A partially-explored region might be matched to a suboptimal analogy, misrepresenting the unobserved part. In that case, multiple reference regions can reduce estimation error and avoid overcommitting to one noisy analogy, provided the analogy error remains below the local sampling error $r_{s,a}(\eta)$.

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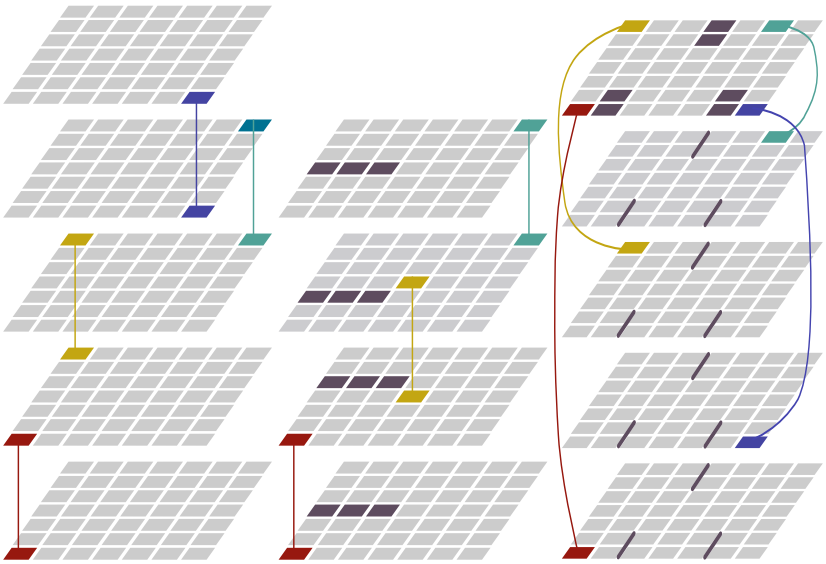
Appendix A: Evaluated domains

All domains are grid-world MDPs with stochastic actions (Fig. 3). Cardinal movement actions succeed with probability 0.9; non-movement actions succeed with probability 0.9 and otherwise no-op. Episodes terminate on goal completion (reward $= 1$) or at horizon H (reward $= 0$).

SequentialFetch is an ordered pickup task with progress $p \in \{0, \ldots, L\}$ tracking collected items; state $s = \langle x, y, p \rangle$; actions $\{\uparrow, \rightarrow, \downarrow, \leftarrow, \textsc{Pickup}\}$.

MultiFloorOffice extends the Office MDP to multiple floors connected by staircases or an elevator; state $s = \langle x, y, f \rangle$; actions $\{\uparrow, \rightarrow, \downarrow, \leftarrow, \textsc{Climb}\}$.

Taxi is stochastic Taxi with passenger location p in state $s = \langle x, y, p \rangle$ and destination treated as the goal; actions $\{\uparrow, \rightarrow, \downarrow, \leftarrow, \textsc{PickUp}, \textsc{DropOff}\}$.



(a) SequentialFetch (b) MultiFloorOffice (stairs) (c) Taxi

Figure 3: **Our domains.** (a) SequentialFetch: The number of layers is determined by the sequence length. (b) MultiFloorOffice (stairs): four floors connected by stairs. A wall shifts along y across floors, disrupting translation invariance. (c) Taxi: $\mathcal{V}$ represents the five different passenger positions (four passenger depots and inside the taxi on the top layer). When the passenger is inside the taxi, walls are dilated by replacing thin walls with blocked cells, yielding a larger collision region.